LIBBY — PATIENT/CAREGIVER TRACK

pancreatic-recurrent-kras-g12r-m8f3

Plain-language summary for patient and caregiver

Generated 2026-05-12

This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

You asked a narrow question. In pancreatic cancer that came back after standard adjuvant chemotherapy, and that carries a KRAS G12R mutation along with a couple of other findings, what targeted options are worth talking about? This page is the patient-facing answer to that question. It walks through one preparation step, then the therapeutic options in rough order of how strong the case for each one is. The clinician-facing version of this page covers the same material with the technical detail your oncologist will want.

One framing thing up front. There has been a lot of news about "KRAS inhibitors" in lung cancer over the last few years, and you may have seen headlines suggesting an off-the-shelf pill now exists. That is true for one specific KRAS variant called G12C. The mutation in your cancer is a different variant called G12R, and the lung-cancer drugs (sotorasib and adagrasib) do not work on G12R. The newer drug class that *does* target G12R-bearing cancers is still in clinical trials. That is why most of the options below are trial enrollments rather than approved prescriptions.

The other framing thing. This page only covers drugs aimed at the specific features in your tumor profile. The standard chemotherapy regimens your oncologist already knows about (FOLFIRINOX, gemcitabine plus nab-paclitaxel, and the rest) are not on this page. That is not because they are wrong. It is because the question was about targeted options, and the chemo conversation is a separate one that lives with your treating team.

What we know about your cancer

You are in your fifties. The cancer started in your pancreas and is the most common pancreatic type (adenocarcinoma). It came back after surgery and adjuvant chemotherapy. The adjuvant regimen was FOLFIRI, an irinotecan-based combination, and the cancer progressed on it. That means you have not yet received a platinum-based chemo (oxaliplatin or cisplatin), which is relevant for one of the options below. You are functional day-to-day with somewhat reduced energy (clinicians score this as ECOG 1). You are not currently on any cancer treatment.

The tumor's molecular profile shows several findings, all confirmed by sequencing:

- A KRAS G12R mutation. KRAS is a gene that, when mutated, keeps cancer cells stuck in "grow" mode. G12R is one of several flavors of this mutation, and it shows up in roughly one in five pancreatic cancers.
- An inactivating mutation in TP53, the most commonly damaged tumor-suppressor gene in human cancer. This is context, not a direct treatment target.
- Loss of a gene called CDKN2A, which normally puts a brake on cell division. When CDKN2A is missing, a nearby gene called MTAP is usually missing too (they sit right next to each other on the chromosome, and they tend to fall out together). Whether MTAP is gone in your case has not been checked yet, and that is one of the questions worth answering up front.
- An alteration in CCND3, a cell-cycle driver that pairs mechanistically with the CDKN2A loss.
- Two immune-system markers that, taken together, make standard immunotherapy unlikely to help. The cancer is "microsatellite stable" (MSS) and has a low tumor mutation burden (4.1 mutations per million bases, which is below the threshold of 10 that the approved immunotherapy regimens require). Immune-checkpoint drugs like pembrolizumab are approved across cancer types only when those markers point the other way. They do not point the other way here, so the immunotherapy class is essentially closed for this cancer at this time. The one exception on this page is an experimental combination that tries to pry that door back open, and even that one

comes with caveats.

What you told us matters most

You said the goal is remission. You said you accept high-risk, high-reward options when the upside is real. You did not flag any specific toxicities you wanted to rule out in advance. You said you prefer clinical trials over off-trial experimental use. Those preferences shaped what got included and what got ranked highly.

The first step regardless of which option you pursue

Before any of the trial-based options below can move forward, the lab work needs to be hardened. This is not optional, but it is also not toxic, and it does not delay treatment by much.

The KRAS G12R call on file came from a local sequencing report. Every trial sponsor running a KRAS-targeting trial re-runs that test on their own central platform at screening, regardless of what the local report says. The same blood draw that confirms the mutation also reads a baseline level of tumor DNA circulating in your bloodstream (called ctDNA). That baseline is useful later. On treatment, a follow-up ctDNA draw at week 6 or 12 can flag whether the cancer is responding weeks before a scan would show it.

Two other tests are worth bundling into this same workup because the same blood draw and the same archived tumor block cover all of them:

- A germline panel. Pancreatic cancer carries a real chance of an inherited gene variant (BRCA1, BRCA2, PALB2, ATM, the Lynch syndrome genes, and a few others) regardless of whether anyone in your family has had cancer. National guidelines now recommend germline testing for every pancreatic-cancer patient. If you carry one of these variants, an entirely new class of drug (PARP inhibitors, discussed below) opens up. If you don't, it stays closed. The result takes three to six weeks. There are no side effects from running the test.
- An MTAP test. This is a quick add-on to the existing tumor sequencing or a separate stain on the same tumor block. It takes one to two weeks. If MTAP is gone (which is likely given the CDKN2A loss), one of the trial options below becomes mechanistically much more interesting.

None of this hurts and none of it commits you to anything. The blood draw plus the tumor-block requisition can be ordered in a single visit, and the answers come back in roughly the time it takes to schedule a trial-screening appointment anyway.

The options

Option 1 — daraxonrasib (RMC-6236), a KRAS-targeted oral drug, in a clinical trial

This is the lead option. It is the one option on this page where there is published clinical data on people with pancreatic cancer carrying a KRAS G12 mutation, taking this specific drug.

What it is: a once-daily pill that targets the KRAS protein in its active state. Unlike the lung-cancer KRAS drugs, this one works across several KRAS variants, including G12R. The trial that has the published readout enrolled 168 people with pancreatic cancer who had received one prior line of treatment. In the subgroup of 26 people dosed at the 300 mg level (the dose this trial is now using), about a third of the cancers shrank measurably, and the median time before the cancer started growing again was around 8.5 months. Median overall survival in that small subgroup was about 13 months. Those numbers are early — the full registration trial (the kind that the FDA usually wants before approval) is still running, and 26 people is a small group to draw firm conclusions from. The G12R version of the mutation was included in the group but has not been reported separately. So the honest answer is: this looks like the best published targeted option for this exact cancer right now, with the caveat that the data are still maturing.

Main side effects: a rash that nearly everyone gets but that is usually manageable with a topical steroid plus a low-dose antibiotic (the same playbook that lung-cancer EGFR-drug patients use), diarrhea in about half of patients, and mouth sores in about a third. About three in ten patients had a serious side effect at some point. No deaths were attributed to the drug.

What fits your preferences: this is a recruiting trial, the side-effect profile is manageable, and the efficacy signal is the strongest in the dossier. The gap against your stated goal of remission is honest — this drug controls the cancer for a meaningful stretch, but the published data do not show people being cured.

Option 2 — PF-07934040, a different KRAS-targeted oral drug, in a clinical trial

This is a backup version of the same general idea. PF-07934040 is a different drug from a different company, with a different chemistry, that also targets KRAS. It is in an earlier phase of testing, which means no efficacy results have been published yet. What makes it worth listing is that the trial specifically names G12R as an eligible variant, which is unusual — most KRAS trials lump all the variants together in the inclusion paperwork, and that can become a screening headache.

The case for this option is essentially: if the first drug above is not accessible (the trial slot is filled, the site is too far, the screening labs disqualify you), this is the next-best version of the same idea.

The risk is that there is no published efficacy data yet. The drug class works in principle, but this specific drug has not yet been shown to work in pancreatic cancer. You would be in an early-phase trial where the goal is partly to figure out how well it works.

The trial does exclude people with significant residual numbness or tingling from prior chemotherapy. Your prior regimen (FOLFIRI) is generally less likely to cause that than a platinum-based regimen would be, but it is worth checking.

Option 3 — daraxonrasib plus a PRMT5 inhibitor (anvumetostat), only if your tumor has lost MTAP

This option targets two of the findings in your tumor at once: the KRAS mutation and the CDKN2A loss. It is gated on the MTAP test mentioned above. Here is why.

MTAP sits next to CDKN2A on the chromosome, and when CDKN2A is lost, MTAP is usually lost too — but not always. If MTAP is lost, then a drug called anvumetostat (which blocks an enzyme called PRMT5 in a way that is selective for MTAP-deleted cells) is mechanistically a very clean fit. The lab data on this drug class are some of the cleanest in cancer biology: MTAP-deleted cancer cells are roughly 70 times more sensitive to the drug than normal cells. The trial in question pairs this drug with daraxonrasib (the same drug as Option 1) on a combination arm, so you get the KRAS coverage and the MTAP-axis coverage in one regimen.

The catches are real. The combination has not yet been shown to work clinically in pancreatic cancer (the bulk of the data so far is in lung cancer and mesothelioma). The side-effect profile of the combination at full dose is still being characterized. And the whole option hinges on MTAP being gone in your tumor — if MTAP is intact, this option is off the table.

Option 4 — olaparib (a PARP inhibitor), only if you carry an inherited BRCA or PALB2 variant *and* a future platinum-chemo response holds up

This option is the only one on the page backed by a randomized phase 3 trial with high-quality evidence. It is also the most heavily gated.

What it is: an oral pill, twice daily, that blocks a DNA-repair enzyme called PARP. In cancers that already have a broken DNA-repair gene (specifically inherited BRCA1, BRCA2, or PALB2 mutations), blocking PARP is a one-two punch that the cancer cells cannot recover from. The pivotal trial enrolled people with metastatic pancreatic cancer who had a germline BRCA mutation and whose cancer was stable after at least four months of platinum-based chemo. Olaparib roughly doubled the time before the cancer started growing again, compared to placebo. The point worth being honest about: the improvement in time-without-progression did not translate (in the published interim analysis) into a measurable improvement in overall survival. Patients lived longer without disease growth, but the data on whether they lived longer overall is so far inconclusive.

Two things have to happen before this option is on the table. First, the germline panel mentioned above has to come back showing one of the relevant variants. The base rate in pancreatic cancer is real but is not the majority — the test is worth running, and the result will be definitive one way or the other. Second, you would need to receive a platinum-containing chemo regimen first (which you have not yet) and have the cancer respond to it for at least four months. That platinum-induction step is a separate conversation with your medical oncologist.

What is contested about this option: it is an approved drug with strong randomized evidence, which is rare on this page, but two structural concerns weigh against ranking it higher. The two gates above are independent and either one can close it. And the survival benefit, as opposed to the time-to-progression benefit, has not been clearly demonstrated. The class also carries a small but real risk of secondary blood cancers (leukemia or myelodysplastic syndrome) with prolonged use, on the order of 1 to 2 percent.

This option carries caveats.

Option 5 — daraxonrasib combined with chemotherapy (FOLFIRINOX or gem/nab) in a clinical trial

This is the same KRAS-targeted drug as Option 1, but added to a standard chemo backbone in a different trial. The case for it: combinations tend to push response rates higher than monotherapy, you have not yet received the kind of chemo most commonly used for metastatic pancreatic cancer (your prior adjuvant regimen was FOLFIRI, which is irinotecan-based but does not include oxaliplatin or platinum), and combining the targeted drug with chemo is consistent with your stated treat-to-remission goal.

The case against ranking it above the monotherapy option: there is a small early-phase abstract (n=6 patients) showing 5 of 6 responding on a different chemo-plus-targeted combination, which is encouraging but is six people in a conference abstract, not a published full paper. There is no published safety table yet for the daraxonrasib-plus-chemo regimen in pancreatic cancer. The side effects of the targeted drug (rash, diarrhea, mouth sores) layer on top of the chemo side effects (low blood counts, fatigue, nerve toxicity), and what that stacked profile looks like at full dose has not yet been reported publicly.

This option carries caveats. It is genuinely an option to discuss, especially given your preferences, but the supporting data are thinner than for Option 1.

Option 6 — zoldonrasib (RMC-7977) plus an immune-system bispecific antibody (ivonescimab)

This is the highest-ceiling, lowest-evidence option on the page. It pairs a related KRAS-targeted drug with a newer immunotherapy that targets two things at once (PD-1, which most immune-checkpoint drugs already hit, plus VEGF, which is involved in tumor blood supply). The reason it is on the page at all is that your tumor's immune profile (MSS, low mutation burden) closes the door on every standard immunotherapy option, and this combination is the only thing on the horizon that tries to reopen that door for pancreatic cancer.

The reason it is not ranked higher: there are zero published clinical results for this combination in pancreatic cancer. The strongest data are from a mouse model of pancreatic cancer that responded dramatically to one of these drugs alone, but that model carried a different KRAS variant (G12D, not G12R), so the extrapolation is two steps removed. The combination's side-effect profile in pancreatic cancer specifically has not been characterized — the rash from the targeted drug and the immune-system inflammation from the bispecific antibody can look similar on the skin, which makes it hard to tell what is causing what. And the trial's listed indications are lung and colorectal cancer; whether the trial has an open pancreatic-cancer slot is something a sponsor inquiry has to confirm.

If you are oriented toward the highest possible ceiling on response and willing to take an option with thin clinical data to get there, this is the lane. If not, it should sit below the better-supported options.

This option carries caveats.

Option 7 — ELI-002, a KRAS-peptide vaccine

This is the cleanest-tolerability option on the page (essentially no serious side effects in the trial data), and the vaccine is built around a panel of seven KRAS variants that explicitly includes G12R. So mechanistically it is a clean fit.

The reason it does not rise higher is that the trial enrolls people in a specific clinical state: pancreatic cancer that was surgically removed and is in remission but with traces of tumor DNA still detectable in the blood (a setting clinicians call "minimal residual disease"). The vaccine is designed to mop up that microscopic residual disease before it grows into something visible. Your cancer is past that window — it has already grown back to a visible recurrence. Future trials may open the indication wider, but the current one does not enroll your clinical state.

This option carries caveats. It is on the page because the mechanism-fit is real and worth knowing about; it is not currently actionable for your specific situation.

Option 8 — avutometinib plus defactinib plus chemotherapy

This is a four-drug combination being tested in pancreatic cancer. There is a small abstract from 2024 showing 5 of 6 patients responding at the lowest dose level, which is striking but is six people and an unpublished abstract. The combination has stronger published data in a different cancer type (low-grade serous ovarian cancer), where it has accelerated FDA approval for a related indication.

The current pancreatic-cancer trial is "active not recruiting," meaning the slots are presently filled. Whether it reopens is the binding question. The combination is mechanistically reasonable but the data supporting it in pancreatic cancer specifically are thin (one conference abstract).

This option carries caveats and is mostly informational at present.

Option 9 — rucaparib (a second PARP inhibitor), under similar gating to Option 4

Rucaparib is in the same drug class as olaparib (Option 4). The reason to list it separately is that its trial enrolled a slightly broader group: instead of only inherited (germline) BRCA mutations, it included people whose BRCA or PALB2 mutation was in the tumor only (somatic), not inherited. So in principle it widens the gate.

The data are weaker — a single-arm phase 2 trial with 46 patients, no comparison group. Median time-without-progression was about 13 months and median survival around 23 months in that small group, but those are secondary readouts in a single-arm study, not registration-grade evidence. The same prerequisite as Option 4 applies: a platinum-chemo induction first, then a response that holds up for at least four months, then maintenance with the drug.

Useful as a backup if Option 4 turns out to be inaccessible.

This option carries caveats.

Option 10 — palbociclib (a CDK4/6 inhibitor) combinations

This option targets the CDKN2A and CCND3 findings in your tumor. Mechanistically, it is the textbook case for a CDK4/6 inhibitor (drugs like palbociclib that are well-established in breast cancer). The lab data combining palbociclib with other targeted drugs in pancreatic-cancer organoids look promising.

The reason it is at the bottom of the list: the closest published clinical evidence is actively negative. A large basket trial that tested palbociclib alone in cancers with this kind of CDK-axis alteration found a response rate of about 4 percent — essentially no signal. Combination approaches that might rescue that signal are at the preclinical stage in pancreatic cancer, not the trial stage. The most relevant combination trial (called ADOPT) is not yet recruiting. Outside of a trial, this would be an off-label combination requiring institutional pharmacy support.

This option carries caveats. It is on the page mostly for completeness — these findings in your tumor have a mechanistic story, but the clinical story to back that story up is not there yet.

Questions to ask your oncologist

- Has the germline panel been ordered yet? When are results expected? If a BRCA or PALB2 result comes back, does it change the immediate plan?
- Is MTAP testable on the existing tumor block, or does it need a new sample? How quickly can it return?
- Among the daraxonrasib trials (Option 1 versus Option 5), which one is geographically accessible, and does prior adjuvant FOLFIRI count as "prior systemic therapy" under the protocol?
- If Option 1 (daraxonrasib monotherapy) is the lead, what is the plan for managing the expected rash? Who is the dermatology contact?

- For the ctDNA monitoring mentioned in the first step — at what intervals on treatment, and what change in the ctDNA level would prompt a change in the plan?
- Has my MMR / microsatellite status been confirmed by IHC in addition to NGS, just to close out the immunotherapy question definitively?
- For Option 4 (olaparib), if germline testing comes back negative, is there value in checking for somatic BRCA / PALB2 (the Option 9 gate) before closing that lane?
- If none of the targeted-trial options is accessible at our center, who is the nearest referral site running these trials, and what does the screening timeline look like there?

Sources

The evidence anchors cited above include:

- Wolpin et al., New England Journal of Medicine 2026 (daraxonrasib in KRAS G12 PDAC) — PMID 42090791
- Holderfield et al., Nature 2024 (preclinical pan-RAS class) — PMID 38589574
- Jiang et al., Cancer Discovery 2024 (preclinical pan-RAS replication) — PMID 38593348
- Wasko et al., Nature 2024 (zoldonrasib in mouse pancreatic cancer model) — PMID 38588697
- Golan et al., New England Journal of Medicine 2019 (POLO trial, olaparib in germline BRCA pancreatic cancer) — PMID 31157963
- Reiss et al., Journal of Clinical Oncology 2021 (rucaparib in BRCA/PALB2 pancreatic cancer) — PMID 33970687
- Mavrakis et al., Science 2016 and Kryukov et al., Science 2016 (PRMT5 / MTAP synthetic lethality) — PMIDs 26912361, 26912360
- Smith et al., Cancer Discovery 2023 (MTA-cooperative PRMT5 selectivity) — PMID 37552839
- Pant et al., Nature Medicine 2024 and Wainberg et al., Nature Medicine 2025 (ELI-002 AMPLIFY-201) — PMIDs 38195752, 40790272
- O'Hara et al., Clinical Cancer Research 2025 (NCI-MATCH Z1C palbociclib basket) — PMID 39437014
- Knudsen et al., Cancer Research 2023 (CDK4/6 combination organoid data) — PMID 36346366

Trial identifiers (for sponsor/registry lookup): NCT05379985, NCT06445062, NCT06447662, NCT06360354, NCT02184195, NCT07397338, NCT05726864, NCT05669482, NCT06813079, NCT06625320.